

Visible Light Communication

From Labs to Market: How Soon?

Overall, the developments are quite interesting, and we could hope to see Li-Fi and other VLC technologies in the mainstream five years down the line

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The radio spectrum is highly congested and the demand for wireless data is making this much worse. Demand for wireless data doubles every year but the available capacity cannot sustain this growth. Even with freeing up of radio frequencies by the switch from old analogue to digital TV, the available capacity is not enough. Offloading of cellular data to Wi-Fi is now a default for smartphones but now the congestion is felt by the Wi-Fi's. Wi-Fi data densities cannot be increased greatly due to interference. So Wi-Fi needs to offload some excess capacity, but to what?" says Dr Gordon Povey, product manager of the University of Edinburgh's D-Light project (which has been funded by Scottish Enterprise), and CEO of PureVLC, a spin-out company formed to commercialise Li-Fi—a new wireless optical communications system developed by them.

Dr Povey explains that squeezing more out of radio technologies or using an alternative such as optical technologies are the only two options available. Regardless of the technology (3G, 4G or Wi-Fi), there are three ways to increase the capacity of wireless radio systems: Find more bandwidth, add more nodes or improve the efficiency of the technology.

In the current situation, more bandwidth is being found but it's clearly not enough—it is finite. More nodes are being added—cell splitting has been done for years—but this is expensive. Also, two nodes do not have double the

